

EDUCATION

- **University of Texas at Austin** Austin, TX
B.S. in Psychology with Honors, Minor in Computer Science; GPA: 3.8 *Aug. 2021 – May. 2025*

EXPERIENCE

- Developing Intelligence Lab** Austin, TX
Lab Technician *May 2025 - Present*

- **Software Development:** Developed MATLAB functions to manipulate, analyze, and visualize behavioral time-series data aligned with research objectives. Maintained the lab's MATLAB repository.
- **Image Embeddings:** Generated ResNet-based embeddings for view-dependent images across 27 categories. Reduced embeddings to two dimensions using multidimensional scaling and applied k-means clustering. Compared unsupervised clusters to ground truth, achieving high alignment.
- **Research Assistant Manager:** Assigned tasks to three undergraduate research assistants and monitored progress to ensure timely completion of computational projects.

- Computational Research Assistant* *Aug 2023 - May 2025*

- **Object Detection for Experimental Studies:** Built a dataset and fine-tuned YOLOv8 using Ultralytics in Python to detect 27 custom objects from egocentric view. Derived time-series features including detection size, distance to center, and category.
- **Software Development:** Developed MATLAB functions to create short videos showing desired behavioral moments from experimental data.

- Data Science Intern* *July 2023 - Aug 2023*

- **Data Analysis of Experiment Results:** Processed raw eye-tracking data into structured features using MATLAB and Python. Applied unsupervised learning (Random Forest, k-means) for participant classification. Presented exploratory findings to lab members.
- **Software Development:** Authored documentation for the lab's MATLAB repository to support research workflows.

PROJECTS

- **Wheeled Robot Pick Up Task Simulation:** Developed motion-planning and control code for a 4-wheeled mobile robot with a 5-DOF arm to pick up and place a cube. Calculated the desired end-effector trajectory and applied a task-space PI control law to obtain the required twist. Computed wheel and joint velocities using the robot Jacobian and a weighted least-squares solution.
- **Controlling NAO through Python:** Built a Python repository leveraging SoftBank Robotics SDK, NumPy, and OpenCV to control actuators, speech, and vision. Implemented higher-level behaviors such as object tracking using image processing and a fine-tuned YOLOv8 model.
- **Randomly Exploring Tree for Path Planning of Kilobot:** Implemented RRT algorithm in Python to compute collision-free paths. Exported the created node graph to CoppeliaSim for motion simulation of Kilobot.

SKILLS

- **Proficient:** MATLAB, Python, Machine Learning, Deep Learning, Image Processing, Git, Numerical Computing, Natural Language Processing, Plotting and Visualizations, Data Structures and Algorithms
- **Comfortable:** C++, Reinforcement Learning, Large Language Models, HTML, SQL,

PUBLICATIONS

Kaplan, B.E, **Martinez, E.**, & Yu, C. (2025). Using Head-Mounted Eye Tracking to Examine Infant Face Looking During Naturalistic Freeplay. Proceedings of the Annual Meeting of the Cognitive Science Society, 47. Retrieved from <https://escholarship.org/uc/item/7z48j6zc>